

Shiheng Zhang
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[Google Scholar](#) | [GitHub](#)

Research Interests

Numerical analysis for high-dimensional and multiscale PDEs; reliable low-rank approximation and time integration; structure-preserving and asymptotic-preserving methods; kinetic and dissipative equations.

Education

Purdue University, West Lafayette, IN 09/2019 – 12/2023

Ph.D. in Computational and Applied Mathematics

Advisor: Jie Shen

Thesis: Energy-dissipative methods in numerical analysis, optimization and deep neural networks for gradient flows and Wasserstein gradient flows

Southern University of Science and Technology, Shenzhen, China 09/2015 – 06/2019

B.S. in Computational and Applied Mathematics

Advisors: Jiang Yang and Bingsheng He

Research Experience

University of Washington, Seattle, WA 01/2024 – Present

Department of Applied Mathematics

Pearson Fellow and Acting Instructor; research with Jingwei Hu

- Dynamical low-rank algorithms for kinetic equations and related PDEs.
- Structure-preserving schemes and scientific machine learning for large-scale computation.

Awards and Honors

- Boeing Research Award, University of Washington, 2025
- T.T. Moh Graduate Scholarship, Purdue University, 2019
- Summa Cum Laude Graduate, Southern University of Science and Technology, 2019
- First Prize, Chinese Mathematics Competitions (Guangdong), Chinese Mathematical Society, 2017

Publications

Full publication list: [Google Scholar](#)

Conference Papers

- C1. **Zhang, S.** *Ordinal geometry complements reconstruction: diagnosing planning with compressed value functions.* **Advances in Neural Information Processing Systems (NeurIPS 2026).**

Journal Articles

- J1. **Zhang, S.** & Hu, J. *On the stability of the low-rank projector-splitting integrator for hyperbolic and parabolic equations.* **Journal of Scientific Computing** 109, 47 (2026).
- J2. **Zhang, S.** & Hu, J. *A separable and asymptotic-preserving dynamical low-rank method for the Vlasov–Poisson–Fokker–Planck system.* **ESAIM: M2AN** 60, 1715–1739 (2026).
- J3. **Zhang, S.**, Shen, J. & Hu, J. *SAV-based entropy-dissipative schemes for a class of kinetic equations.* **SIAM Journal on Scientific Computing** (2025).
- J4. Einkemmer, L., Hu, J. & **Zhang, S.** *Asymptotic-preserving dynamical low-rank method for the stiff nonlinear Boltzmann equation.* **Journal of Computational Physics** (2025).
- J5. **Zhang, S.** & Shen, J. *Structure preserving schemes for a class of Wasserstein gradient flows.* **Communications on Applied Mathematics and Computation** (2025).
- J6. **Zhang, S.**^{*}, Zhang, J.^{*}, Shen, J. & Lin, G. *A relaxed vector auxiliary variable algorithm for unconstrained optimization problems.* **SIAM Journal on Scientific Computing** (2025). (*Equal contribution).
- J7. Zhang, J.^{*}, **Zhang, S.**^{*}, Shen, J. & Lin, G. *Energy-dissipative evolutionary deep operator neural networks.* **Journal of Computational Physics** (2024). (*Equal contribution).

Preprints

- R1. **Zhang, S.** & Hu, J. *Robust high-order projector-splitting integrators.* arXiv: [2608.17157](#).
- R2. **Zhang, S.** *A counterexample to robust second-order convergence of the Strang projector-splitting integrator.* arXiv: [2608.12893](#). Submitted.
- R3. **Zhang, S.** *Guidance breaks the fitted operator: a terminal-fitted repair for classifier-free guidance.* arXiv: [2607.07665](#). Submitted.
- R4. **Zhang, S.** *Asymptotic preservation and uniform accuracy of diffusion and flow-matching samplers.* arXiv: [2607.04113](#).
- R5. **Zhang, S.** & Shen, J. *SAV schemes with decomposition-induced pullback corrections for gradient flows.* arXiv: [2606.18551](#). Submitted.
- R6. **Zhang, S.** *Planner-admissible graph-PDE value extensions for sparse goal-conditioned planning.* arXiv: [2605.19185](#). Submitted.
- R7. Zhang, J., **Zhang, S.** & Lin, G. *Low-rank adaptation of evolutionary deep neural networks for efficient learning of time-dependent PDEs.* arXiv: [2509.16395](#). Submitted.

Talks and Posters

Talks

- T1. **Zhang, S.** *Robust projector-splitting Runge–Kutta integrators of orders two and three.* Minisymposium talk, SIAM Conference on Computational Science and Engineering (CSE27), Pittsburgh, PA. 2027.
- T2. **Zhang, S.** *On the stability of the low-rank projector-splitting integrator for parabolic equations.* Talk, CHaRMNET Annual Meeting, University of Washington. 2025.
- T3. **Zhang, S.** *Structure preserving schemes for a class of Wasserstein gradient flows.* Invited minisymposium talk, SIAM Annual Meeting, Spokane, WA. 2024.
- T4. **Zhang, S.** *A relaxed vector auxiliary variable algorithm for unconstrained optimization problems.* Invited minisymposium talk, SIAM Conference on Optimization, Seattle, WA. 2023.

Posters

- P1. **Zhang, S.** *A separable and asymptotic-preserving dynamical low-rank method for the Vlasov–Poisson–Fokker–Planck system.* Poster, IPAM, UCLA, Los Angeles, CA. 2026.
- P2. **Zhang, S.** *SAV-based entropy-dissipative schemes for a class of kinetic equations.* Poster, ICERM, Providence, RI. 2025.
- P3. **Zhang, S.** *Asymptotic-preserving dynamical low-rank method for the stiff nonlinear Boltzmann equation.* Poster, CHaRMNET Annual Meeting, Virginia Tech. 2024.
- P4. **Zhang, S.** *SAV-based entropy-dissipative schemes for a class of kinetic equations.* Poster, NSF Computational Mathematics PI Meeting, University of Washington. 2024.

Conference Organization

Co-organizer, Minisymposium: *Low-Rank and Tensor Methods for High-Dimensional PDEs*, SIAM Conference on Computational Science and Engineering (CSE27), Pittsburgh, PA, February 2027

Teaching

Instructor, University of Washington

- AMATH 351 A and B: Introduction to Differential Equations and Applications – Spring 2026
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2026
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2025
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2025
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2024

Teaching Assistant, Purdue University

Courses: MA 265: Linear Algebra; MA 266: Ordinary Differential Equations; MA 261: Multivariate Calculus; MA 162: Calculus II; MA 161: Calculus I